

Hubs, Bridges & Climate Discourse: Analysing User Interaction and Community Structure in Reddit Climate Discussions

COSC2671 / COSC3047 - Social Media and Network Analytics - Assignment 2

Dataset: cathw/reddit_climate_comment (HuggingFace, MIT Licence)

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Table of Contents

1. Introduction.....	3
2. Background and Related Work.....	4
3. Data.....	5
3.1 Source and Collection.....	5
3.2 Preprocessing and Cleaning.....	5
3.3 Limitations.....	5
4. Methodology.....	6
4.1 Sentiment Analysis (VADER).....	6
4.2 TF-IDF Keyword Analysis.....	6
4.3 Topic Modelling (LDA).....	6
4.4 Network Construction.....	7
4.5 Network Analysis Measures.....	7
5. Results and Discussion.....	8
5.1 Dataset Overview.....	8
5.2 Sentiment Analysis Results.....	10
5.3 TF-IDF Keyword Analysis.....	10
5.4 Topic Modelling Results.....	11
5.5 User Interaction Network.....	13
5.6 Community Detection.....	15
5.7 Subreddit Networks.....	15
6. Conclusion.....	17
7. References.....	18

1. Introduction

Climate change is one of the most widely debated topics on social media, and Reddit provides a uniquely rich environment for studying how these discussions form and spread across online communities. Reddit is organised around both users and topic-specific communities (subreddits), enabling participants to reply to each other, form communities, and develop distinct discourse norms. Climate change conversations on Reddit range from scientific discussion and policy debate to scepticism, activism, and everyday concerns about energy, waste, and sustainability.

This report investigates the following research question:

"How do Reddit users interact in climate change discussions, and which users, topics, and subreddit communities act as hubs or bridges in the climate discourse network?"

This question is operationalised through six explicit success criteria, each answered in the Conclusion (Section 6):

- SC1: Identify the main thematic frames using LDA topic modelling with k selected via a perplexity evaluation sweep.
- SC2: Demonstrate statistically significant sentiment differences across subreddits using the Kruskal-Wallis non-parametric test.
- SC3: Construct a directed, weighted user interaction network and identify influential users via PageRank and approximate bridge score.
- SC4: Detect communities using greedy modularity maximisation and interpret the structure in terms of discourse organisation.
- SC5: Build a bipartite subreddit-topic network connecting subreddits to dominant LDA topics (threshold $\geq 15\%$ of messages).
- SC6: Build a subreddit similarity network using TF-IDF cosine similarity and identify thematic bridge communities.

The `cathw/reddit_climate_comment` dataset (Wang, 2024) was selected because it contains genuine Reddit usernames and a complete post-comment-reply hierarchy, enabling a true directed user interaction network - more difficult with fully anonymised alternatives.

2. Background and Related Work

Climate change discussions on Reddit are shaped by interactions between users and communities. Reddit allows users to participate across many subreddits, creating a large discussion space where information, opinions, and arguments can spread between groups. Some communities focus on science and environmental concern, while others discuss climate change through politics, economics, scepticism or technology.

Research in computational social science has consistently shown that online discussions form distinct communities with limited cross-community exchange. Bail et al. (2018) found that exposure to opposing views on social media can paradoxically increase polarisation rather than reduce it. Williams et al. (2015) specifically examined Reddit climate communities, finding they occupy distinct ideological positions with limited cross-community information flow, creating echo chambers. Olson and Neal (2015) demonstrated that Reddit subreddits sharing users tend to cluster by topic and ideology.

Natural language processing methods are widely used to analyse social media text at scale. Latent Dirichlet Allocation (Blei et al., 2003) uncovers hidden thematic structure by modelling each document as a mixture of latent topics without requiring labelled data. VADER sentiment analysis (Hutto & Gilbert, 2014) was specifically designed for social media text, handling capitalisation, degree modifiers, and emoticons natively. TF-IDF analysis reveals community-specific vocabulary by weighting terms frequent within a community but rare across others.

Network analysis provides a structural perspective. PageRank (Page et al., 1999) identifies users whose content important others engage with, going beyond raw interaction counts. Bridge-based measures can identify users who appear to connect different parts of the network, acting as possible information conduits. Community detection using greedy modularity maximisation (Clauset et al., 2004) identifies user clusters who interact more with each other than with outsiders. Modularity $Q > 0.3$ indicates meaningful community structure; $Q > 0.5$ indicates strong structure (Newman, 2006).

3. Data

3.1 Source and Collection

The dataset is `cathw/reddit_climate_comment`, hosted on HuggingFace under the MIT Licence (Wang, 2024). It was collected using the PRAW Python library via the Reddit API on 21 and 22 February 2024 across eight climate-focused subreddits: `r/climate`, `r/climatechange`, `r/energy`, `r/environment`, `r/renewableenergy`, `r/sustainability`, `r/climateactionplan`, and `r/zerowaste`. The raw dataset contains 6,550 posts in a nested parquet format (post -> comments -> replies), flattened to 113,127 rows before deduplication. After preprocessing, the final dataset comprised 96,328 interactions. The full dataset (112 MB) exceeds the 10 MB submission limit; a representative stratified sample (10,620 rows, 8.83 MB) is included with the submission. No API keys or private credentials are included in any submitted files.

Field	Type	Role in NLP Analysis	Role in Network Analysis
CommentBody / ReplyBody / post_body	Text	Primary input for VADER sentiment, LDA topic modelling, TF-IDF	Body text for topic assignment in bipartite network
CommentAuthor / ReplyAuthor	String	Authorship metadata	Source nodes in directed user interaction graph
post_author	String	Authorship metadata	Target nodes for comment on post edges
subreddit	String	Stratification for per-community NLP analysis	Community grouping for subreddit-level networks
CommentUpvotes / post_upvotes	Integer	Engagement weighting	Engagement score computation
CommentTimestamp / post_timestamp	DateTime	Temporal ordering	Ordering of interaction events

Table 1. Dataset fields and their explicit role in NLP and network analysis.

3.2 Preprocessing and Cleaning

The nested parquet structure was flattened into a row-per-interaction format. A directed edge list was built simultaneously: `comment_author` -> `post_author` for each comment; `reply_author` -> `comment_author` for each reply. Body text was unified into a single `clean_text` field. Rows with missing IDs, missing authors, or text under 2 characters were removed. Duplicate messages were deduplicated by message ID. Preprocessing applied: lowercasing, URL removal, HTML entity normalisation, user/subreddit mention removal, non-alphanumeric character stripping, whitespace normalisation, and stopword handling during vectorisation. Deleted accounts (`[deleted]`), bots (AutoModerator), and self-loop edges were removed before network construction, eliminating 728 of 83,518 raw edges (0.87%).

3.3 Limitations

The dataset was collected over a short two-day period on 21 and 22 February 2024, which limits temporal generalisability. Eight pre-selected subreddits may not represent all Reddit climate discourse. VADER has documented difficulty with sarcasm and irony common in Reddit comments. Because exact betweenness centrality was computationally prohibitive on the full 48,936-node network, an approximate bridge score using sampled shortest paths was used. LDA was trained on a 30,000-row stratified sample for computational efficiency, which may introduce sampling variability.

4. Methodology

All analysis was implemented in Python using: pandas and NumPy (data processing); NLTK SentimentIntensityAnalyzer for VADER sentiment analysis; scikit-learn CountVectorizer, LatentDirichletAllocation, TfidfVectorizer, and cosine_similarity (topic modelling and similarity); scipy.stats (Kruskal-Wallis); and NetworkX (network construction, centrality, and community detection). The approach was explicitly iterative: the number of LDA topics was determined empirically via a perplexity sweep, and the similarity threshold was selected by inspecting graph connectivity.

4.1 Sentiment Analysis (VADER)

VADER was selected over supervised classifiers because: (1) it requires no labelled Reddit-specific training data; (2) its lexicon was designed for social media text, handling capitalisation, punctuation, and degree modifiers natively; and (3) it is computationally efficient for a 96,328-text corpus (Hutto & Gilbert, 2014). The compound score (-1 to +1) was classified as positive (≥ 0.05), negative (≤ -0.05), or neutral. The Kruskal-Wallis H test was applied to confirm statistical significance of sentiment differences - the appropriate non-parametric alternative to ANOVA when distributions are skewed and normality cannot be assumed.

4.2 TF-IDF Keyword Analysis

TF-IDF was computed per subreddit using a 5,000-term vocabulary (max_df=0.90, bigrams included). TF-IDF weights terms frequent within a community but rare across communities, revealing community-specific vocabulary. This complements LDA topic modelling by providing a lexical fingerprint for each subreddit independent of the topic model structure.

4.3 Topic Modelling (LDA)

Latent Dirichlet Allocation was chosen over word clouds or frequency analysis because it captures co-occurrence patterns, modelling each document as a mixture of latent topics (Blei et al., 2003). A CountVectorizer was applied with max_df=0.85, min_df=5, max_features=6,000, and ngram_range=(1,2). The model was fitted and interpreted on a 30,000-row sample for computational efficiency. The number of topics k was selected empirically via a perplexity sweep across k in {4, 5, 6, 7, 8, 10}. $k=8$ achieved the lowest perplexity (2,298.83), followed closely by $k=7$ (2,302.09). However, lower perplexity did not produce the most interpretable final structure. After inspecting the top words, $k=6$ was selected because it produced clearer and less fragmented topic groups for the final analysis.

k	Perplexity	Log-likelihood	Decision
4	2,361.54	-4,393,722	Baseline; broader topics
5	2,373.73	-4,396,634	Higher perplexity
6	2,360.40	-4,393,449	SELECTED: clearer interpretable topics
7	2,302.09	-4,379,300	Lower perplexity but more fragmented
8	2,298.83	-4,378,498	Lowest perplexity; less interpretable
10	2,353.27	-4,391,737	Over-specified / fragmented

Table 2. LDA perplexity evaluation sweep. $k=8$ had the lowest perplexity; $k=6$ was selected for interpretability.

4.4 Network Construction

Network 1 - Directed User Interaction Graph

A directed weighted graph $G1 = (V, E)$ where V is the set of Reddit users. A directed edge ($A \rightarrow B$) with weight w exists if user A commented on a post by B (`comment_on_post`) or replied to a comment by B (`reply_to_comment`), with weight equal to the aggregated interaction count. The graph is directed because commenting has a clear sender and receiver - this asymmetry encodes who initiates engagement versus who receives it. The graph is weighted to capture repeated interactions as stronger ties. Deleted accounts and bots were removed before construction (728 edges removed).

Network 2 - Subreddit Similarity Graph

Each subreddit was represented as a TF-IDF vocabulary vector. Cosine similarity was computed for all 28 subreddit pairs. Cosine similarity measures vocabulary shape independently of community size, making it appropriate for comparing thematic framing. An undirected weighted graph $G2$ was built including edges where cosine similarity ≥ 0.50 , a threshold retaining all 8 nodes while filtering genuinely dissimilar pairs.

Network 3 - Bipartite Subreddit-Topic Graph

An undirected bipartite graph $G3 = (S \cup T, E)$ where S is the set of 8 subreddits and T is the set of 6 LDA topics, with edges only between the two sets. An edge (s, t) with weight w exists if subreddit s assigns $\geq 15\%$ of its messages to topic t , filtering marginal associations. This network directly encodes which communities engage with which thematic frames. The bipartite property was formally verified using `nx.is_bipartite() = True`.

4.5 Network Analysis Measures

PageRank ($\alpha=0.85$) identifies users engaged with by other high-PageRank users - distinguishing genuine authority from raw activity. Weighted in-degree identifies popular targets; weighted out-degree identifies active producers. An approximate bridge score using sampled shortest paths was used as a computationally feasible proxy for betweenness centrality, identifying users who structurally connect different discourse clusters. Community detection used greedy modularity maximisation (Clauset et al., 2004) on the undirected user graph. Modularity $Q > 0.3$ indicates meaningful community structure (Newman, 2006). For computational feasibility, PageRank, approximate bridge score, and community detection were calculated on a top-2,000 degree-node analysis subgraph, while overall structural statistics were reported for the full user interaction network.

5. Results and Discussion

5.1 Dataset Overview

After flattening, deduplication, and cleaning, the final dataset comprised 96,328 interactions across 8 subreddits. Figure 1 shows the distribution of messages across communities.

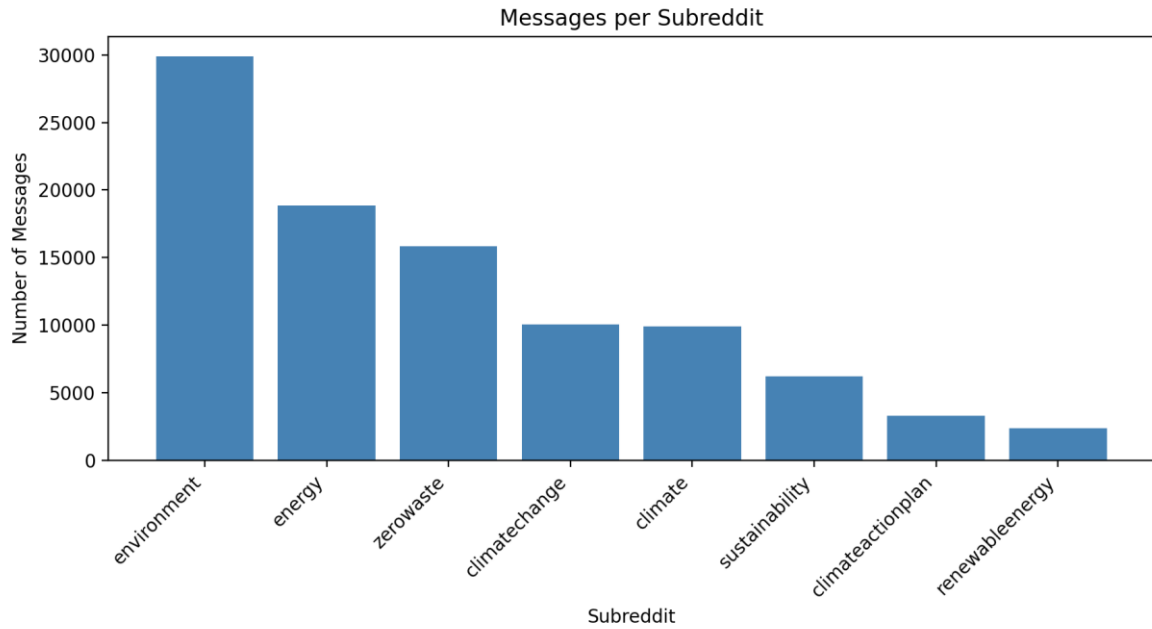


Figure 1: Distribution of messages across subreddits. *r/environment* is the largest community (29,878 messages, 31.0%). *r/renewableenergy* is the smallest (2,354 messages, 2.4%).

Subreddit	Messages	% of corpus
r/environment	29,878	31.0%
r/energy	18,831	19.5%
r/zerowaste	15,829	16.4%
r/climatechange	10,032	10.4%
r/climate	9,922	10.3%
r/sustainability	6,213	6.4%
r/climateactionplan	3,269	3.4%
r/renewableenergy	2,354	2.4%
Total	96,328	100%

Table 3. Interaction counts by subreddit.

r/environment dominates with 31.0% of all interactions, reflecting its broad scope. The ten most active users were mafco (881 messages), silence7 (265), ILikeNeurons (224), Splenda (193), mutatron (187), Tpain63 (133), fungussa (126), Speculawyer (126), FANGO (125), and Ericus1 (125). This long-tail distribution is characteristic of online social networks.

5.2 Sentiment Analysis Results

VADER sentiment analysis was applied to all 96,328 body texts. Figure 2 shows the overall sentiment distribution and mean compound scores by subreddit.

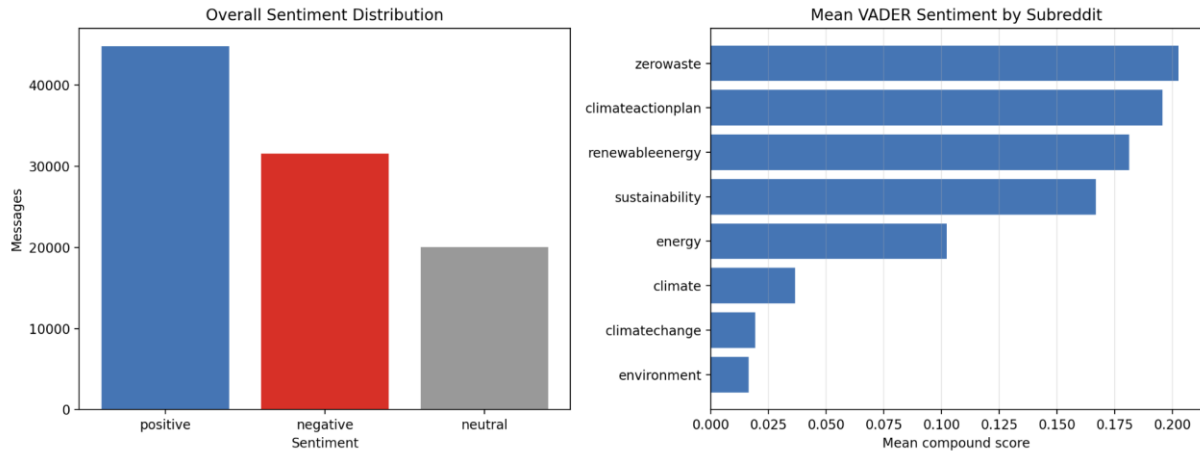


Figure 2: Sentiment analysis results. Left: corpus is net positive (44,771 positive / 31,545 negative / 20,012 neutral). Right: solution-focused communities show the highest mean sentiment, while broader crisis-focused communities are less positive.

The corpus is net positive: 44,771 positive (46.5%), 31,545 negative (32.7%), and 20,012 neutral (20.8%). The Kruskal-Wallis test confirms that sentiment differences across subreddits are statistically significant ($H = 1984.44$, $p < 0.001$).

Subreddit	Mean compound	Positive %	Character
r/zerowaste	0.203	~55%	Most positive - lifestyle/community framing
r/climateactionplan	0.196	~57%	Positive - solution and action-oriented
r/renewableenergy	0.181	~54%	Positive - technology optimism
r/sustainability	0.167	~53%	Positive - constructive framing
r/energy	0.102	~48%	Mixed - fossil fuel and energy debate
r/climate	0.037	~42%	Less positive - urgency and crisis framing
r/climatechange	0.019	~42%	Less positive - scientific risk discourse
r/environment	0.016	~42%	Lowest mean sentiment - broad environmental crisis

Table 4. Sentiment mean by subreddit. Kruskal-Wallis $H=1984.44$, $p<0.001$.

A clear pattern emerges: communities framing climate change through solutions and action generate more positive sentiment than communities using crisis-oriented and scientific-risk framing. This supports the interpretation that community framing shapes the emotional tone of climate discourse.

5.3 TF-IDF Keyword Analysis

TF-IDF keyword analysis revealed community-specific vocabulary providing a lexical fingerprint for each subreddit, complementing the LDA topic assignments.

Subreddit	Top TF-IDF terms (community-specific vocabulary)
r/climate	questions concerns, message compose, subreddit message, automatically contact, compose questions, moderators subreddit, contact moderators, action performed
r/climateactionplan	forest, evs, nuclear power, plant based, olivine, reactors, vegan, planted
r/climatechange	warmer, sea level, ice age, glaciers, level rise, el, atmospheric, ppm

r/energy	evs, heat pump, solar wind, nuclear power, inflation, putin, ercot, ukraine
r/environment	vegan, obama, fucked, vegetarian, bags, forest, bottles, plant based
r/renewableenergy	evs, gw, mwh, wind turbines, solar wind, mw, energy storage, green hydrogen
r/sustainability	vegan, plant based, bags, protein, clothes, packaging, oat, vegetarian
r/zerowaste	bags, zero waste, reusable, bag, bottles, cup, packaging, soap

Table 5. Top TF-IDF terms by subreddit. Vocabulary unique to each community relative to the corpus.

r/renewableenergy uses specialist technical vocabulary such as mwh, gw, wind turbines, energy storage, and green hydrogen. r/zerowaste and r/sustainability share lifestyle vocabulary (vegan, bags, packaging). r/climatechange uses scientific measurement vocabulary (ppm, glaciers, sea level). r/environment also contains political and activist vocabulary, suggesting overlap between environmental and political discourse.

5.4 Topic Modelling Results

Figure 3 shows the LDA perplexity evaluation across candidate topic counts, justifying the selection of k=6.

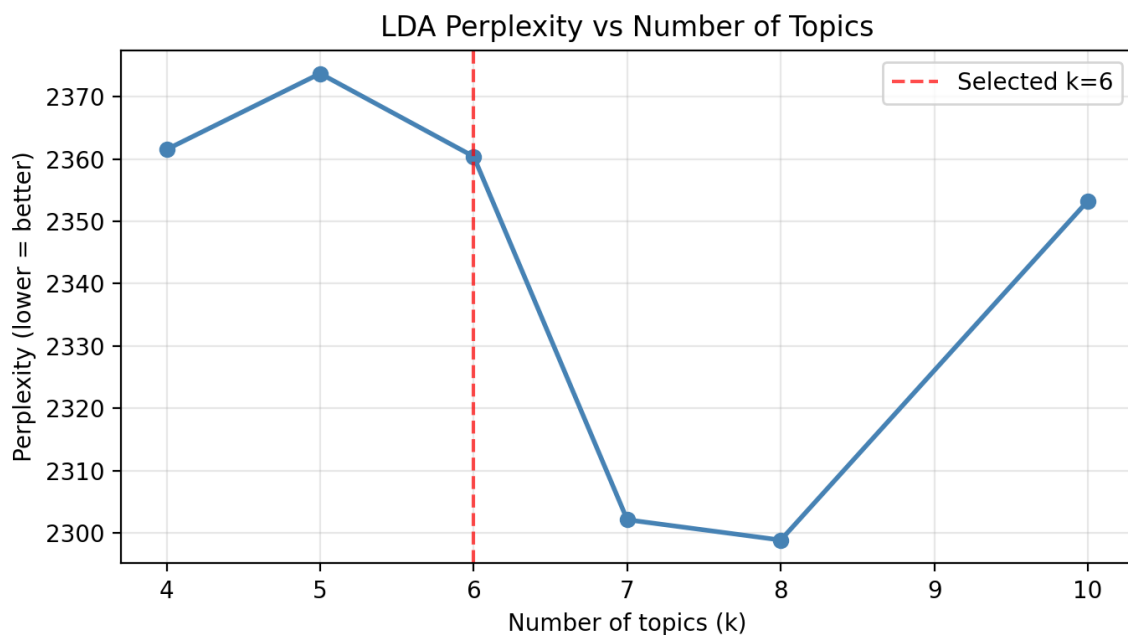


Figure 3: LDA perplexity sweep. k=8 had the lowest perplexity, while k=6 was selected for clearer and more interpretable final topic groups.

k=8 achieved the lowest perplexity (2,298.83), followed closely by k=7 (2,302.09). However, topic selection was not based on perplexity alone. After inspecting the topic words, k=6 was selected because it produced clearer and less fragmented topic groups for the final analysis. Table 6 presents the final model based on the 30,000-row LDA sample, and Figure 4 shows the topic distribution within that sample.

Topic	Label	Top keywords	Count
0	Energy Systems and Technology	energy, solar, power, nuclear, wind, coal, grid	3,620

1	Water, Waste and Heat	water, just, like, heat, waste, don, use	4,314
2	Fossil Fuels and Emissions	gas, oil, carbon, emissions, fossil, years, fuel	3,732
3	Plastic Waste and Consumer Behaviour	use, plastic, like, just, ve, don, really	4,468
4	Lifestyle, Food and Public Discussion	people, like, don, just, meat, think, food	5,028
5	Climate Science and Global Impact	climate, change, people, climate change, just, don, like	8,838

Table 6. LDA topic labels (k=6) from the current notebook output.

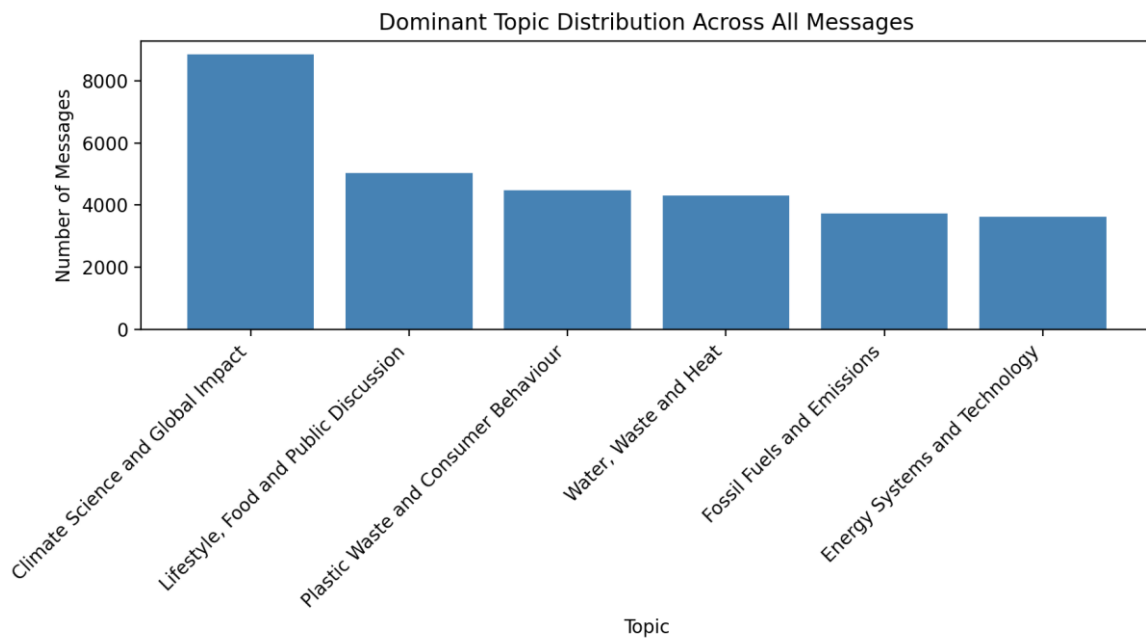


Figure 4: LDA topic distribution. 'Climate Science and Global Impact' is largest, followed by 'Lifestyle, Food and Public Discussion'.

Climate Science and Global Impact is the largest topic in the 30,000-row LDA sample, followed by Lifestyle, Food and Public Discussion. The topic distribution shows that Reddit climate discourse includes both broad science/policy discussion and everyday lifestyle or consumption themes. No single topic dominates all communities, suggesting that climate discourse is thematically fragmented.

5.5 User Interaction Network

The directed user interaction graph was constructed from 82,790 clean edges. Table 7 shows the structural properties.

Property	Value	Interpretation
Nodes (users)	48,936	Unique Reddit users in climate discourse
Directed edges	81,270	Unique directed user-user interaction pairs
Density	3.39×10^{-5}	Extremely sparse - typical of large online social networks
Avg. clustering coefficient	0.018	Low local clustering; most interactions are dyadic
Reciprocity	0.085 (8.5%)	Most interactions are one-directional, justifying directed design
Largest weakly connected comp.	48,149 (98.4%)	Near-complete connectivity across all users

Table 7. Structural properties of the directed user interaction network.

The low reciprocity (8.5%) is structurally significant: most interactions are one-directional, consistent with a broadcast-style participation pattern and justifying the directed graph design. Figure 5 shows the top users by PageRank and Figure 6 shows the top bridge users.

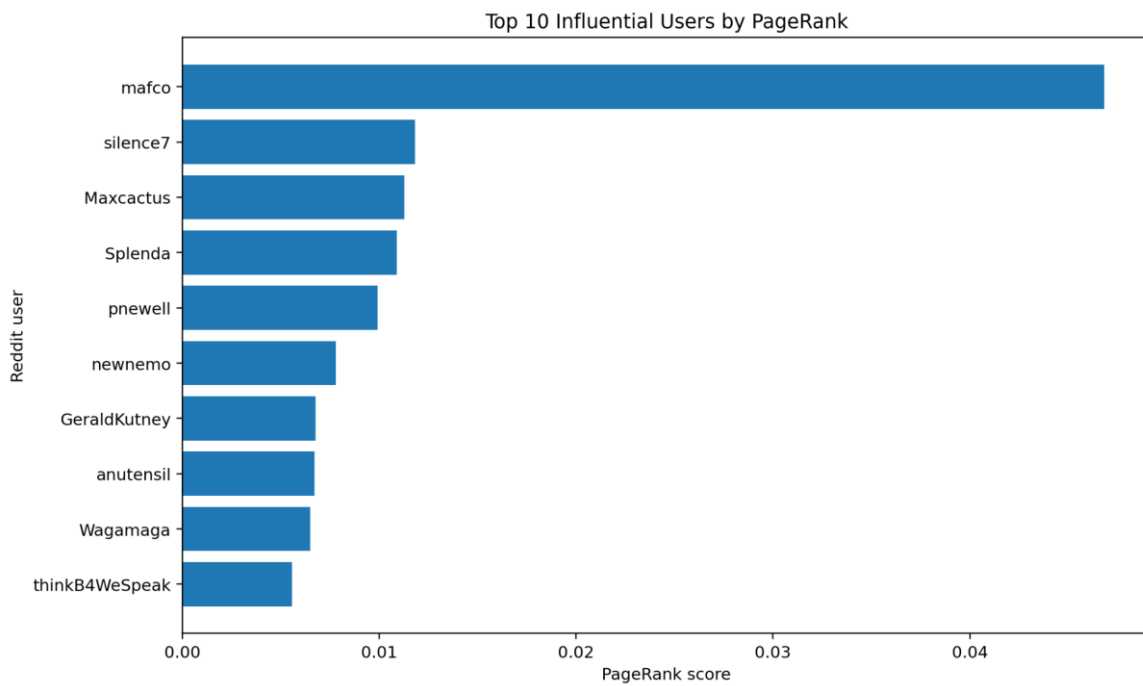


Figure 5: Top users by PageRank). mafco (PageRank=0.047) dominates, 4.7x the next user (silence7=0.013). PageRank identifies authoritative users whose content important others engage with.

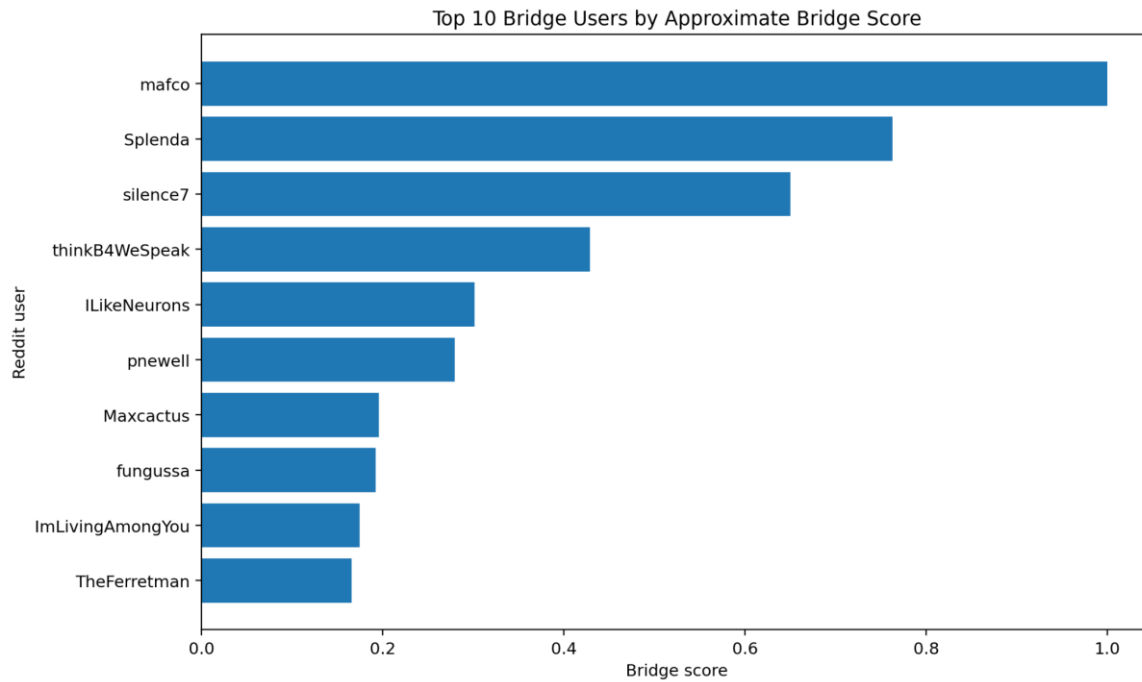


Figure 6: Top bridge users by approximate bridge score . mafco, Splenda, and silence7 are authority-bridges with both high PageRank and high bridge scores. ILikeNeurons and thinkB4WeSpeak are bridge users with lower PageRank than the main authority-bridges.)

User	PageRank	In-degree	Out-degree	Bridge score	Role
mafco	0.047	788	224	1.000	Dominant hub and bridge
silence7	0.012	192	52	0.650	Authority-bridge
Maxcactus	0.011	146	7	0.196	Authority; rarely initiates
Splenda	0.011	180	92	0.763	Authority-bridge
pnewell	0.010	90	4	0.280	Quiet authority
ILikeNeurons	0.004	45	80	0.302	Pure bridge
thinkB4WeSpeak	0.006	52	38	0.429	Bridge user
TheFerretman	0.006	40	45	0.166	Active bridge

Table 8. Top user centrality profiles from the notebook output. Bridge score is an approximate connector measure.

User mafco is the dominant hub: PageRank 0.047, weighted in-degree 788, weighted out-degree 224, and bridge score 1.000. Two bridge archetypes emerge. Authority-bridges such as mafco, Splenda, and silence7 combine high PageRank with high bridge scores, while users such as ILikeNeurons and thinkB4WeSpeak mainly help connect parts of the interaction network.

5.6 Community Detection

82 communities were detected with modularity $Q = 0.5121$, above the $Q > 0.3$ threshold for meaningful structure (Newman, 2006). Table 9 profiles the largest communities.

Community	Size	Top PageRank users	Key observation
Community 0	468	pnewell, newnemo, anutensil, Wagamaga, thinkB4WeSpeak	Contains major authority/bridge users
Community 1	408	mafco, Maxcactus, Splenda,	Contains major authority/bridge users

		TheFerretman, duke_of_alinor	
Community 2	339	silence7, GeraldKutney, YoanB, ILikeNeurons, jayclaw97	Contains major authority/bridge users
Community 3	313	Infinite-Mud3931, boppinmule, antihostile, Molire, shallah	Large interaction cluster
Community 4	207	ImLivingAmongYou, wowhahafuck, koolspaz2, aconsideredlife, Tinkerbellhair	Large interaction cluster

Table 9. Largest user communities. $Q=0.5121$, 82 total communities.

Communities span interactions across multiple subreddits, suggesting that users cluster by interaction patterns and shared discussion spaces rather than subreddit membership alone. Top-authority users are distributed across different communities. The high modularity confirms strong clustering, suggesting that users tend to interact more within their own groups than across groups. Bridge users may help connect these clusters.

5.7 Subreddit Networks

5.7.1 Subreddit Similarity Network

Figure 7 shows the TF-IDF cosine similarity network at threshold 0.50.

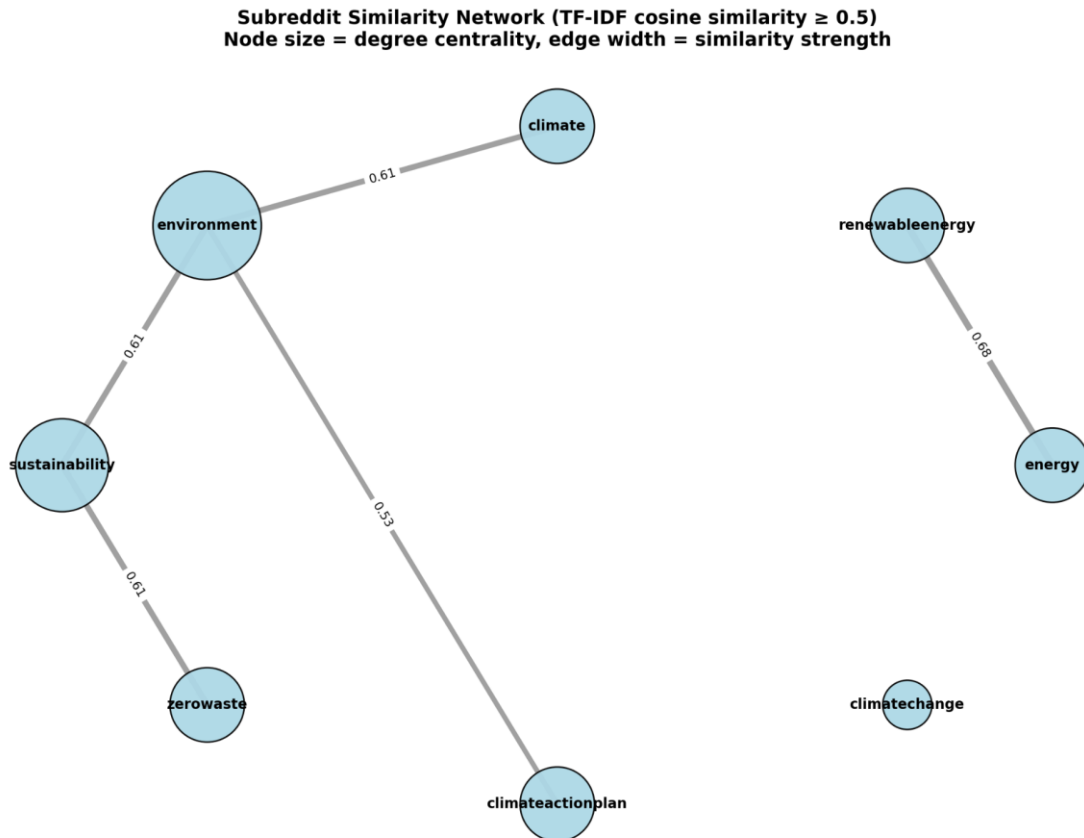


Figure 7: Subreddit similarity network. Cosine similarity ≥ 0.50 ; 8 nodes, 5 edges. $r/environment$ is most connected (degree 0.43). $r/climatechange$ has no edges at this threshold (degree 0.00).

Subreddit pair	Cosine similarity	Interpretation
energy - renewableenergy	0.680	Shared energy technology vocabulary
climate - environment	0.614	Shared science and crisis vocabulary
environment - sustainability	0.612	Shared environmental lifestyle vocabulary
sustainability - zerowaste	0.611	Shared consumption and lifestyle vocabulary
climateactionplan - environment	0.531	Policy and activism overlap

Table 10. Top subreddit similarity pairs. Network captures lexical similarity (TF-IDF), complementing the bipartite network which captures thematic similarity (LDA).

r/environment has the highest degree centrality (0.43) in the similarity network. r/climatechange has degree 0.00 at this threshold - its specialist scientific vocabulary (ppm, glaciers, sea level) is sufficiently distinct that it shares no pair above 0.50, confirming it as the most lexically isolated community.

5.7.2 Bipartite Subreddit-Topic Network

Figure 8 shows the bipartite subreddit-topic network.

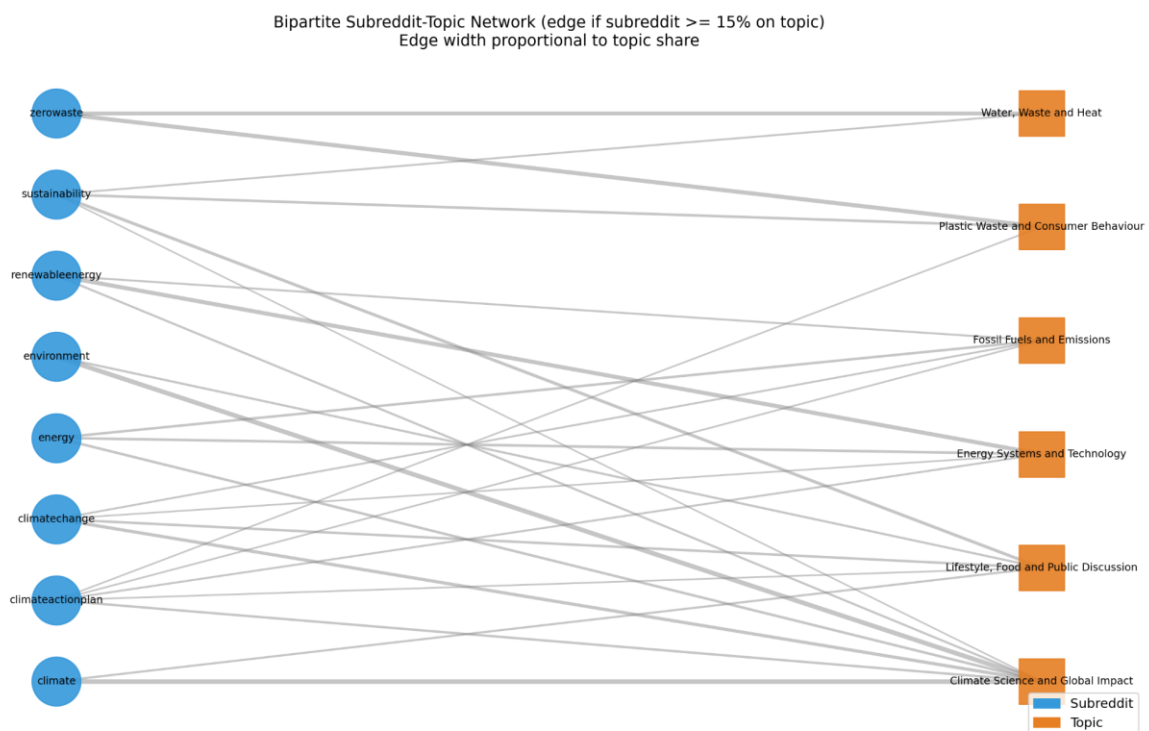


Figure 8: Bipartite subreddit-topic network. The graph contains 14 nodes and 25 edges. An edge exists if a subreddit allocates at least 15% of its sampled messages to a topic.

Subreddit	Most connected topics (>15%)	Key finding
r/climate	Climate Science and Global Impact (44%); Lifestyle, Food and Public Discussion (20%)	Diverse thematic range
r/climateactionplan	Climate Science and Global Impact (23%); Energy Systems and Technology (19%); Fossil Fuels and Emissions (17%); Lifestyle, Food and Public Discussion (16%); Plastic Waste and Consumer Behaviour (16%)	Action and policy oriented

r/climatechange	Climate Science and Global Impact (31%); Lifestyle, Food and Public Discussion (24%); Fossil Fuels and Emissions (18%); Energy Systems and Technology (16%)	Science and public discussion focus
r/energy	Energy Systems and Technology (25%); Climate Science and Global Impact (24%); Fossil Fuels and Emissions (24%)	Spans clean and fossil fuel frames
r/environment	Climate Science and Global Impact (44%); Lifestyle, Food and Public Discussion (21%)	Broad bridge across frames
r/renewableenergy	Energy Systems and Technology (39%); Climate Science and Global Impact (21%); Fossil Fuels and Emissions (19%)	Energy-specific community
r/sustainability	Lifestyle, Food and Public Discussion (29%); Plastic Waste and Consumer Behaviour (25%); Water, Waste and Heat (18%); Climate Science and Global Impact (15%)	Lifestyle and sustainability framing
r/zerowaste	Plastic Waste and Consumer Behaviour (40%); Water, Waste and Heat (36%)	Lifestyle and waste-focused

Table 11. Bipartite subreddit-topic associations above the 15% threshold. bipartite=True verified in the notebook.

All 6 topics connect to at least one subreddit, confirming all identified frames are genuinely present. The Plastic Waste and Consumer Behaviour topic connects to multiple subreddits- the most cross-community thematic frame. Energy Systems and Technology has its strongest connection to r/renewableenergy - the most community-specific frame. r/energy spans both energy technology and fossil fuel/emissions frames, reflecting its ideologically diverse community.

6. Conclusion

This study analysed 96,328 Reddit interactions across eight climate-related communities to understand how climate change is discussed and how users connect within the wider discourse network. The results show that Reddit climate discussion is not one single conversation, but a mix of different themes, emotional tones, influential users, bridge users, and community clusters.

- SC1 - Six distinct topical frames were identified using LDA on a 30,000-row sample, with $k=6$ selected for interpretability after the perplexity sweep: Climate Science and Global Impact (8,838), Lifestyle, Food and Public Discussion (5,028), Plastic Waste and Consumer Behaviour (4,468), Water, Waste and Heat (4,314), Fossil Fuels and Emissions (3,732), and Energy Systems and Technology (3,620). No community is monothematic.
- SC2 - Sentiment differences are highly statistically significant (Kruskal-Wallis $H=1984.44$, $p<0.001$). Corpus is net positive (46.4%). Solution-focused communities (r/climateactionplan: ~57% positive; r/zerowaste: ~55%) are substantially more positive than crisis-framing communities (r/environment: mean 0.016; r/climatechange: mean 0.019).
- SC3 - Directed user interaction network (48,936 nodes, 81,270 edges, density 3.39×10^{-5} , reciprocity 8.5%). mafco is the dominant hub (PageRank 0.047, 4.7x next user, in-degree 788). Two bridge archetypes: authority-bridges (mafco, Splenda, silence7) and bridge users (ILikeNeurons, thinkB4WeSpeak).
- SC4 - 82 user communities detected ($Q=0.5121$, meaningful structure). Top-authority users distributed across different communities - no single community monopolises influential voices. Strong clustering patterns were confirmed.
- SC5 - Bipartite subreddit-topic network (14 nodes, 25 edges, bipartite verified). Plastic Waste and Consumer Behaviour is a cross-community frame; Energy Systems and Technology is most community-specific (r/renewableenergy dominant).
- SC6 - Subreddit similarity network (8 nodes, 5 edges, threshold=0.50). r/environment is most central bridge community (degree 0.43). r/climatechange is most lexically isolated (degree 0.00).

These findings have practical implications for climate communication. Communities that frame climate change as solvable through technology, lifestyle, or policy generate substantially more positive sentiment. The strong clustering structure ($Q=0.5121$, 82 communities) indicates that targeting bridge users - both authority-bridges and pure bridges - may be more effective for cross-community information diffusion than targeting high-PageRank authorities already embedded within specific clusters.

Limitations include the short two-day collection period, eight pre-selected subreddits, VADER's sarcasm limitations, the approximate bridge score, and LDA training on a 30,000-row sample. Future work could extend to longer time series, deploy BERTopic for richer topic modelling, and implement exact betweenness centrality where computationally feasible.

7. References

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